Vehicle Parking Management System

System Validation Report

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Executed 8 November 2026 on commit 39e0e56. 748 of 748 Jest tests passed in 39 suites.

1. Purpose

This report validates the finished system against the SRS: every functional requirement through its System Test case, every non-functional requirement through the tests that measure it, and the main user flows end to end in real browsers.

2. How the system was validated

- System test cases (ST-xx_06 in the Test Plan) run against the full API and a real MySQL database.
- Browser tests (Playwright, e2e/) start the real server on the parking_test database and drive the built web client in Chrome, Edge and Firefox: registration, vehicle registration, entry, slot view, tracking, moving a vehicle, exit, UPI failure then cash payment, history, admin slots/rates/users, and access control.
- Load and concurrency tests seed 500 slots, 5,000 vehicles and 100 users and act at the same moment.
- Windows 11, Node.js 24.22.0, MySQL 8 (parking_test database, rebuilt from database/schema.sql before the run), Jest 30 with Istanbul coverage, Playwright 1.64.

3. Functional requirements: system test cases

Case	Req.	Description	Expected	Actual	Result
ST-01_06	FR-01	Lock account after 5 failed logins	Account should be temporarily locked (SEC-04).	5th wrong password locked the account with a message showing the time left.	Pass
ST-02_06	FR-02	Updated vehicle persists after re-login	Updated colour should be displayed.	Blue still shown after logging out and in again.	Pass
ST-03_06	FR-03	Availability response time	Page should load within 2 seconds for 95% of requests (NFR-01).	95th percentile under 2 seconds over 20 requests with 300 slots (NFR-01).	Pass
ST-04_06	FR-04	Reservation shown in occupancy report	Occupancy count should include the reservation.	Occupancy report counted the reservation.	Pass
ST-05_06	FR-05	Entry completes within time limit	Entry should complete within 3 seconds (NFR-02).	Entry completed in well under 3 seconds (about 0.1 s) (NFR-02).	Pass
ST-06_06	FR-06	Slot shows Occupied everywhere	Slot should show Occupied on every screen.	Occupied for driver, attendant and admin, in Track vehicle and in the report.	Pass
ST-07_06	FR-07	Tracking shows latest slot	The new slot should be shown, not the old one.	New slot shown, not the old one, after the attendant moved the vehicle with Move vehicle (#102).	Pass
ST-08_06	FR-08	Completed session cannot be edited	Edit should be blocked (DR-10).	No route edits a session; a second exit is refused (DR-10).	Pass
ST-09_06	FR-09	Fee shown without delay	Fee should be shown within the 3-second exit flow (NFR-02).	Fee returned in well under 3 seconds (about 0.04 s) (NFR-02).	Pass
ST-10_06	FR-10	Payment linked to session	Payment should be linked to the correct session (SYS-13).	Payment linked to the correct session, for that session's fee (SYS-13).	Pass
ST-11_06	FR-11	History with many sessions	All records should be reachable without errors.	Paging reached every one of the 50+ sessions once, no errors.	Pass
ST-12_06	FR-12	Audit log records admin actions	Log should show the action, administrator and time (SEC-07).	Log showed the action, the administrator and the time.	Pass

All Test Plan cases per functional requirement

Req.	Requirement	Cases passed	Status
FR-01	Register an account and log in using valid credentials.	6/6	Validated
FR-02	Register, view, update and manage vehicle information.	6/6	Validated
FR-03	View the current availability and status of parking slots.	6/6	Validated
FR-04	Reserve an available parking slot for a specified period.	6/6	Validated
FR-05	Record vehicle entry and create an active parking session.	6/6	Validated
FR-06	Assign an appropriate available slot and prevent duplicate assignments.	5/6	Validated with deviation
FR-07	Identify the current parking location and status of a parked vehicle.	6/6	Validated
FR-08	Record vehicle exit and complete the parking session.	6/6	Validated
FR-09	Calculate parking duration and charges from the configured rates.	6/6	Validated
FR-10	Make and record parking payments.	6/6	Validated
FR-11	Maintain and display historical parking-session information.	6/6	Validated
FR-12	Manage users, vehicles, slots and rates, and generate parking reports.	6/6	Validated

FR-06 deviation: IT-06_05 (attendant overrides the assigned slot) was not built. Slot assignment is automatic; an attendant can move a parked vehicle to another slot afterwards (#102, verified by ST-07_06 and the browser test).

4. Non-functional requirements

NFR	Requirement	Tests	Evidence	Passed	Status
NFR-01	Slot availability shown within 2 s for 95% of requests.	slot-availability.test.js; scalability.test.js	p95 under 2 s over 20 requests with 300 slots (ST-03_06), and for 100 users at once viewing 500 slots.	35/35	Validated
NFR-02	Entry and exit processed within 3 s.	vehicle-entry.test.js; vehicle-exit.test.js	Entry about 0.1 s (ST-05_06); fee at exit about 0.04 s (ST-09_06).	35/35	Validated
NFR-03	99% availability during operating hours.	reliability.test.js	Health check reports app and database; 99% of 600 mixed requests at 20 at a time answered without a server error; a database error gives a clean 500 and the next request works.	18/18	Validated
NFR-04	500 slots, 5,000 vehicles, 100 concurrent users.	scalability.test.js; move-vehicle.test.js	Database seeded with 500 slots and 5,000 vehicles; 100 users log in, list slots, list vehicles and park at the same moment, each getting a different slot.	38/38	Validated
NFR-05	Attendant completes a normal entry within 1 minute.	e2e/attendant-entry-usability.spec.js (Chrome, Edge; Firefox in CI)	Login to slot on screen in about 2-3 s in the browser; one-field, keyboard-only entry; 10 entries in a row. See also NFR-05-usability-test.md.	10/10	Validated
NFR-06	Consistent slot and session data under simultaneous operations.	reliability.test.js; scalability.test.js	Same vehicle entering 10 times at once gets one session; 30 bikes for 10 slots: exactly 10 parked, no slot given twice.	26/26	Validated
NFR-07	Modular components for users, vehicles, parking, payment and reporting.	Code inspection	Checked by inspection: server/src/modules has separate users, vehicles, parking, payments and reports modules (plus slots, rates, reservations, history, audit, admin), each with routes, controller, service, repository and validation files.	-	Validated (inspection)
NFR-08	Works in Chrome, Edge and Firefox.	e2e/main-flows.spec.js (Chrome, Edge; Firefox in CI); e2e/attendant-entry-usability.spec.js (Chrome, Edge; Firefox in CI)	All 12 browser tests pass in each of Chrome, Edge and Firefox.	24/24	Validated
NFR-09	Accurate relationships between users, vehicles, slots, sessions and payments.	reliability.test.js	Integrity checker finds no orphan or mismatched rows after concurrent entries, exits and payments.	18/18	Validated
NFR-10	Authentication required for protected functions.	authorization.test.js; session-timeout.test.js; data-protection.test.js	Every protected route refuses a request without a session (401); the route inventory test fails if a new route is added without a rule.	184/184	Validated

NFR	Requirement	Tests	Evidence	Passed	Status
NFR-11	Different privileges for Drivers, Attendants and Administrators.	authorization.test.js; data-protection.test.js; admin-users.test.js; audit-log.test.js	Each role gets 403 outside its role, e.g. a driver tracking a vehicle (IT-07_05).	211/211	Validated
NFR-12	Passwords stored with secure hashing.	password-security.test.js; tests/unit/password.test.js	Only a 60-character bcrypt hash is stored; same password gives different hashes; the hash cannot be used to log in.	14/14	Validated
NFR-13	Account locked after 5 consecutive failed logins.	lockout.test.js	5th failure locks with a message showing the time left; lock ends after the period (ST-01_06).	11/11	Validated
NFR-14	Inactive sessions end after 15 minutes.	session-timeout.test.js	Idle 14 min still works; idle 15 or 16 min refused with 401 and deleted.	12/12	Validated
NFR-15	Input validated before processing or storing.	validation.test.js; tests/unit/*.validation.test.js	Every field error reported with a clear message; malformed JSON gives 400, oversized body 413.	176/176	Validated
NFR-16	Audit log of important admin activities.	audit-log.test.js	Rate, slot, user and vehicle changes are logged with action, admin and time (ST-12_06).	10/10	Validated
NFR-17	Only administrators can read audit logs.	audit-log.test.js	Attendant and driver refused with 403 (IT-12_03).	10/10	Validated
NFR-18	No unauthorised access to user and payment data.	data-protection.test.js; payments.test.js	Users cannot reach other users' data; full card number and expiry never stored, returned or logged.	178/178	Validated

"Passed" counts every test in the listed suites, including tests that check other requirements in the same file.

5. Browser tests (NFR-05, NFR-08)

Flow	Chrome	Edge	Firefox
attendant-entry-usability.spec.js: 10 vehicles in a row: each entry is quick, and the field is empty and ready for the next one	Passed	Passed	Passed
attendant-entry-usability.spec.js: a mistake is easy to fix: an unknown number shows a message, keeps the text, and the right number then works	Passed	Passed	Passed
attendant-entry-usability.spec.js: a normal entry from the login page to the slot on screen takes far less than 1 minute	Passed	Passed	Passed
attendant-entry-usability.spec.js: pressing Enter on an empty field says what to do	Passed	Passed	Passed
attendant-entry-usability.spec.js: the entry screen needs one field and works from the keyboard alone	Passed	Passed	Passed
main-flows.spec.js: UPI fails at the exit and the attendant takes cash later from the waiting list (UC-08, FR-10)	Passed	Passed	Passed
main-flows.spec.js: a driver sees a finished stay in parking history (UC-11)	Passed	Passed	Passed
main-flows.spec.js: a new driver registers, logs in and registers a vehicle (UC-01, FR-02)	Passed	Passed	Passed
main-flows.spec.js: an administrator adds a slot, opens the rates page and adds a user (UC-12)	Passed	Passed	Passed
main-flows.spec.js: an attendant moves a parked vehicle and Track vehicle shows the new slot (#102, FR-07)	Passed	Passed	Passed
main-flows.spec.js: an attendant records entry, sees the slot occupied, finds the vehicle and records exit (UC-05, UC-03, UC-07, UC-08)	Passed	Passed	Passed
main-flows.spec.js: pages are protected: a driver cannot open admin pages, and log out ends the session (NFR-10, NFR-11)	Passed	Passed	Passed

Chrome and Edge: run locally on 8 November 2026 on the same application code as commit 39e0e56 (later commits change only Jest tests), 24 of 24 passed. Firefox: the Playwright Firefox build is blocked from starting on the local Windows machine ("spawn UNKNOWN"), so the Firefox results come from GitHub Actions run 41589654563 on main (3 Nov 2026, after #104 merged), where all 36 browser tests (12 per browser) passed on Ubuntu. The commits since then add only tests.

6. Conclusion

All 12 functional requirements are met. 71 of 72 Test Plan cases pass; the remaining case (IT-06_05) is a documented deviation with a replacement feature. All 18 non-functional requirements are validated: 17 by automated tests and NFR-07 (modular structure) by inspection of the code structure.